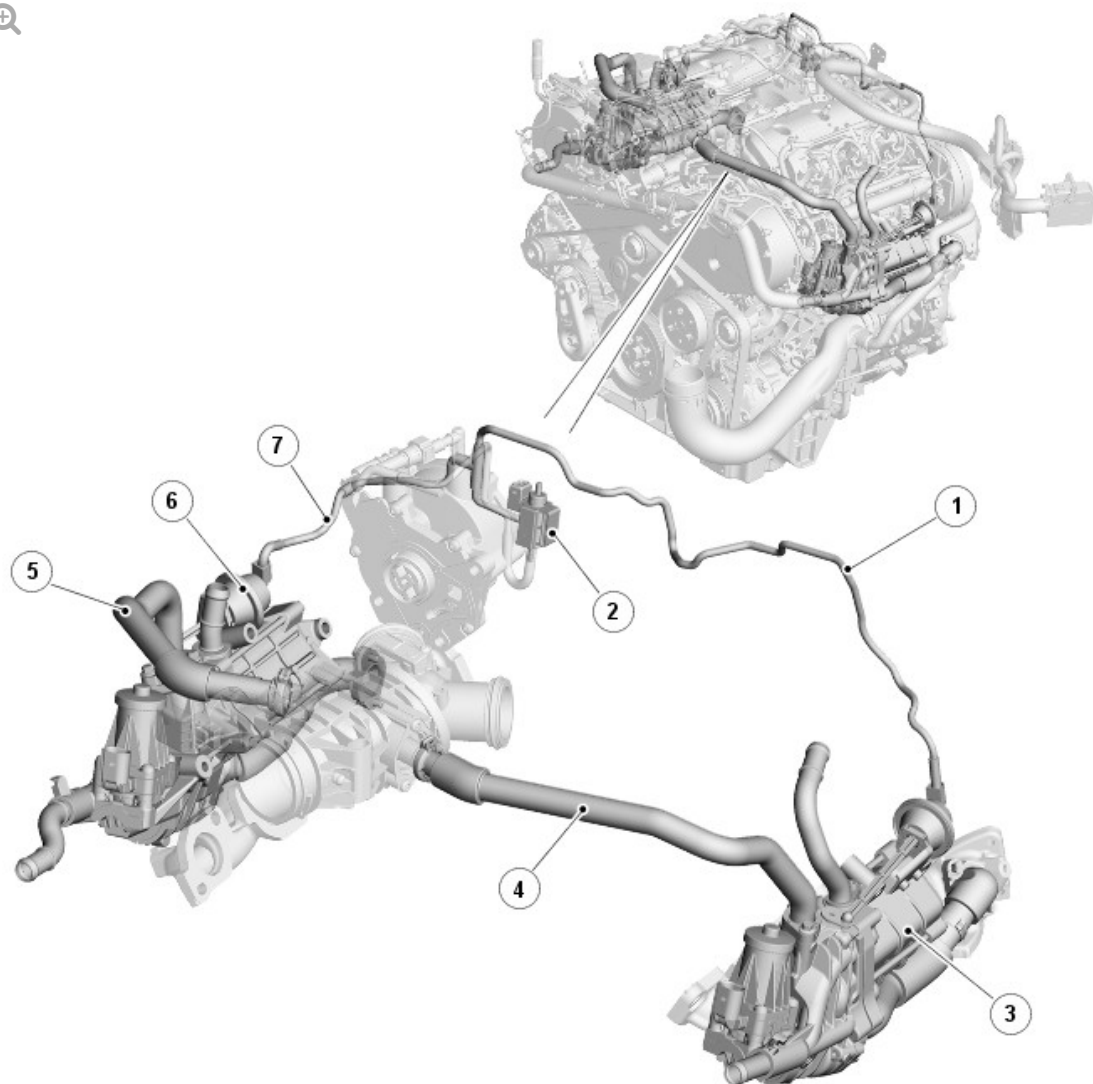


2016.0 RANGE ROVER (LG), 303-08

ENGINE EMISSION CONTROL - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

DESCRIPTION AND OPERATION

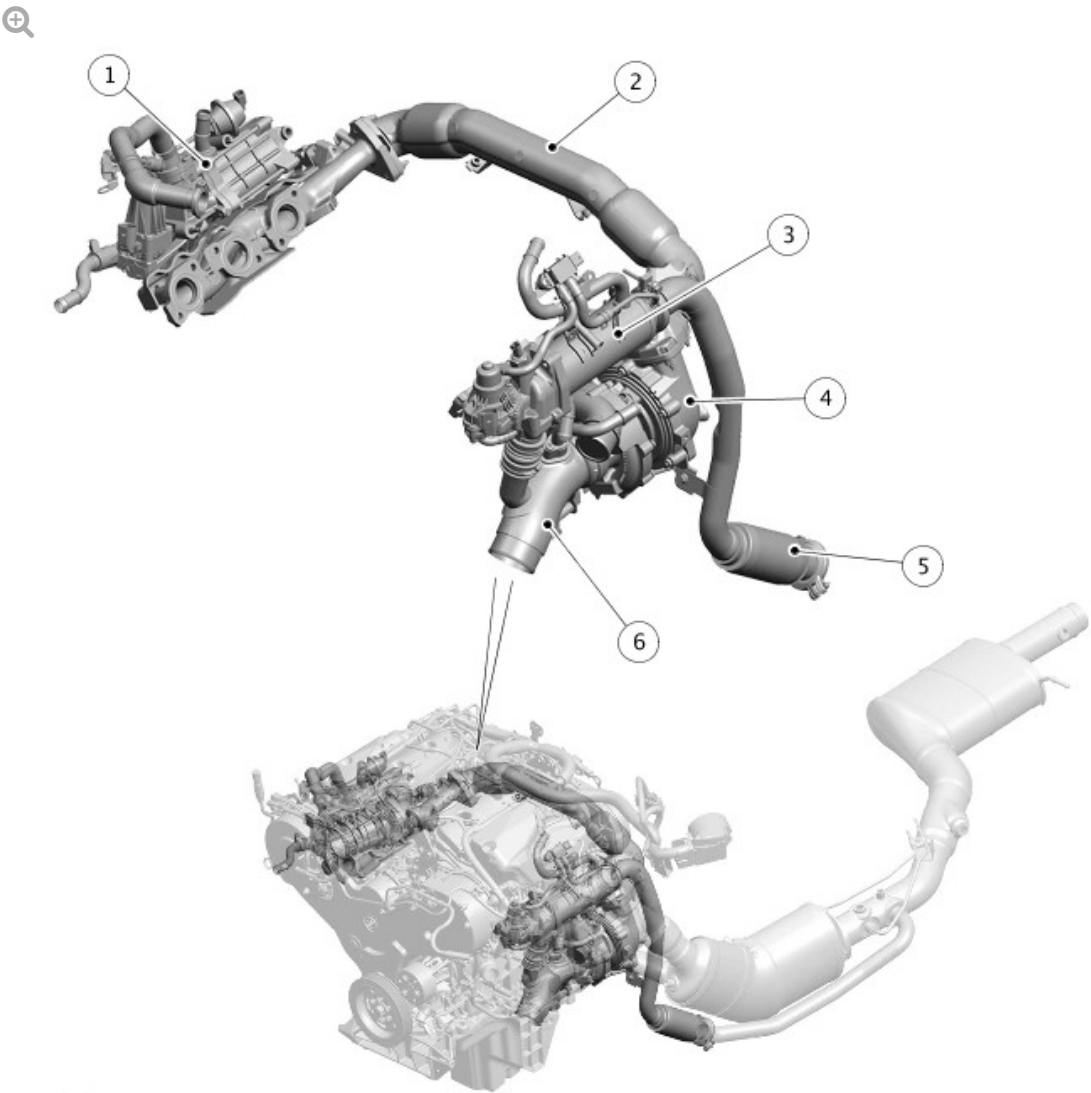
COMPONENT LOCATION - EXHAUST GAS
RECIRCULATION



E146140

ITEM	DESCRIPTION
1	Left vacuum tube
2	EGR cooler bypass valve solenoid
3	Left EGR cooler and valve housing
4	Left EGR outlet tube
5	Right EGR outlet tube
6	Right EGR cooler and valve housing
7	Right vacuum tube

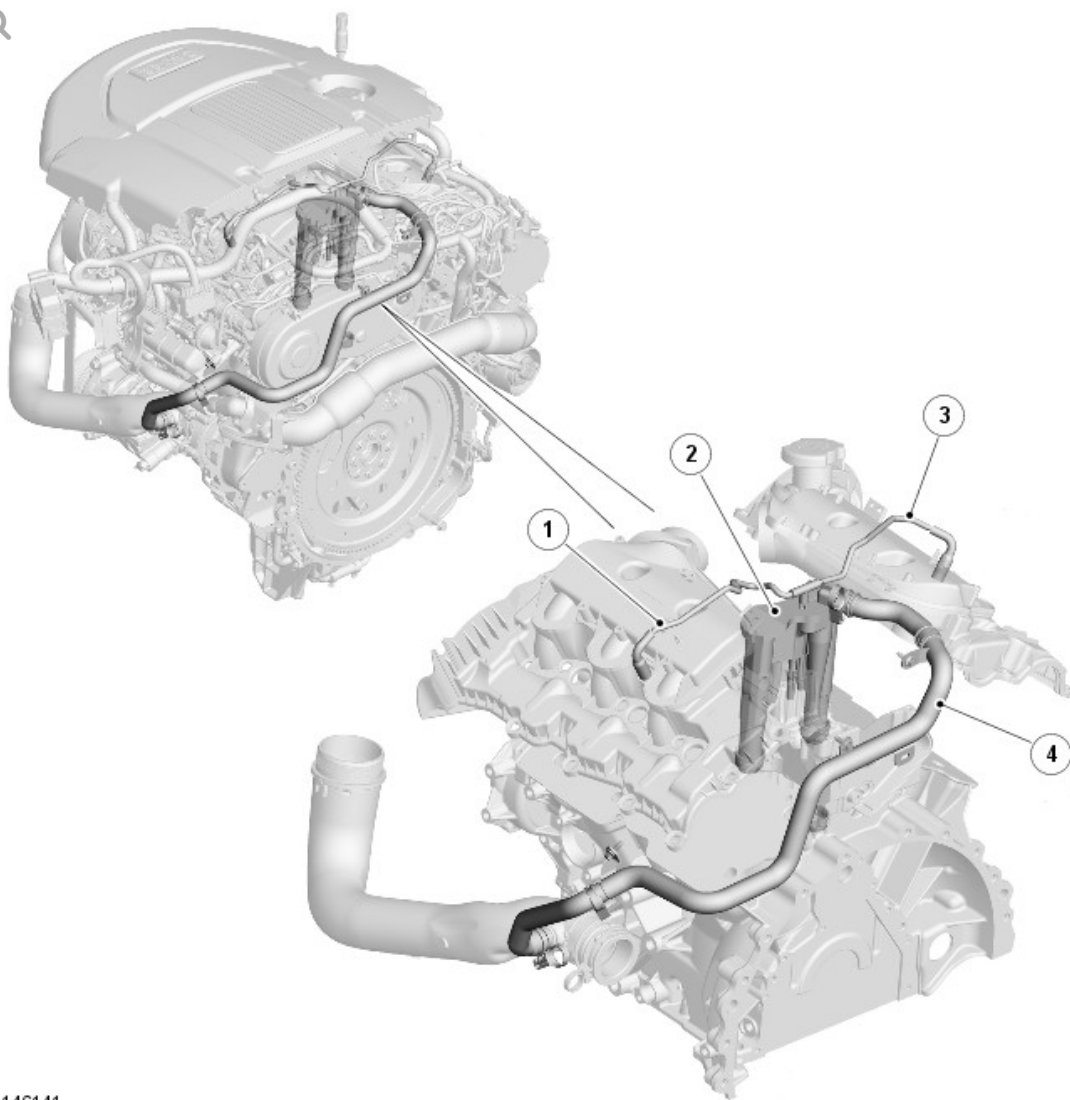
COMPONENT LOCATION - LOW
PRESSURE (LP) EXHAUST GAS
RECIRCULATION SYSTEM



E171872

ITEM	DESCRIPTION
1	Right EGR cooler and valve housing
2	Exhaust manifold
3	Low Pressure (LP) Exhaust Gas Recirculation (EGR) cooler and valve housing
4	Turbocharger
5	Exhaust gas inlet from post Diesel Particulate Filter (DPF)
6	Clean air inlet from air intake system

COMPONENT LOCATION - CRANKCASE VENTILATION



E146141

ITEM	DESCRIPTION
1	Left camshaft cover ventilation tube
2	Oil separator
3	Right camshaft cover ventilation tube
4	Crankcase ventilation hose and tube assembly

OVERVIEW

EXHAUST GAS RECIRCULATION SYSTEM

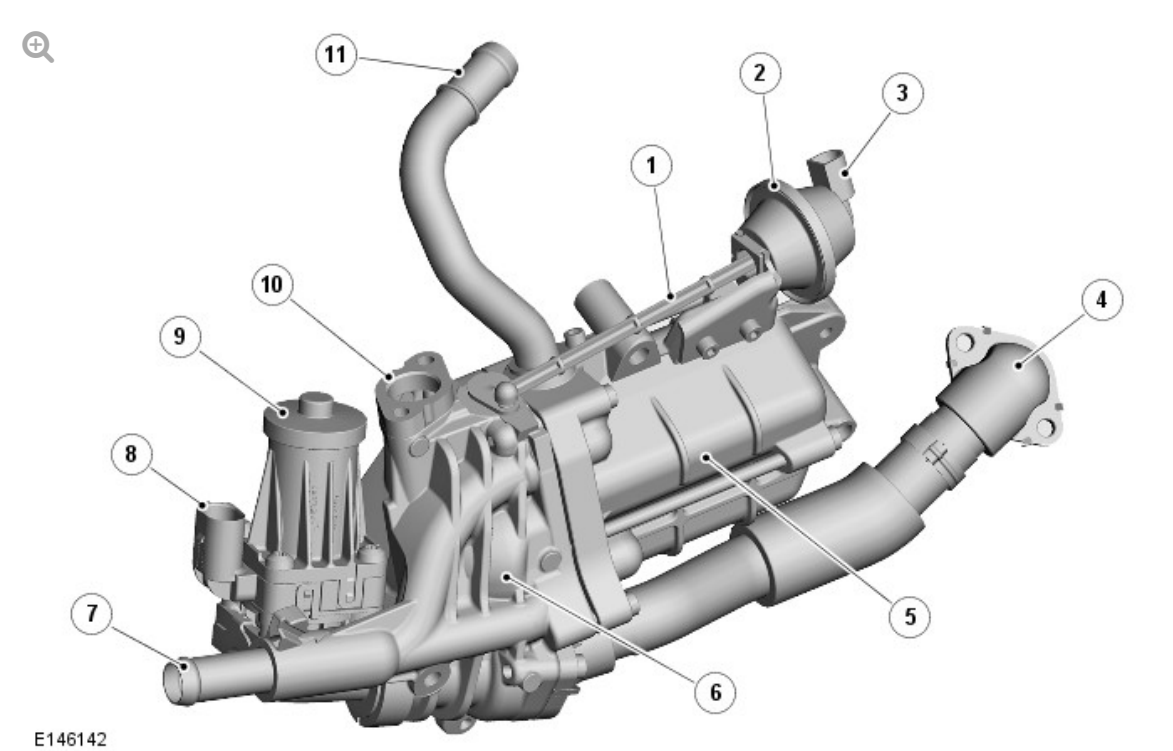
The EGR system is used to control the amount of exhaust gas being recirculated in order to reduce exhaust emissions and combustion noise. EGR operation is calibrated and could operate at sub-zero ambient temperature as well as transient conditions.

CRANKCASE VENTILATION SYSTEM

The crankcase ventilation system ensures that all gasses emitted from the crankcase during engine running are separated from any oil particles and recirculated via the clean air induction system. It will also stop buildup of pressure inside the crankcase thereby ensuring integrity of engine seals

DESCRIPTION

EXHAUST GAS RECIRCULATION SYSTEM



ITEM	DESCRIPTION
1	Actuator rod
2	Cooler bypass valve vacuum actuator

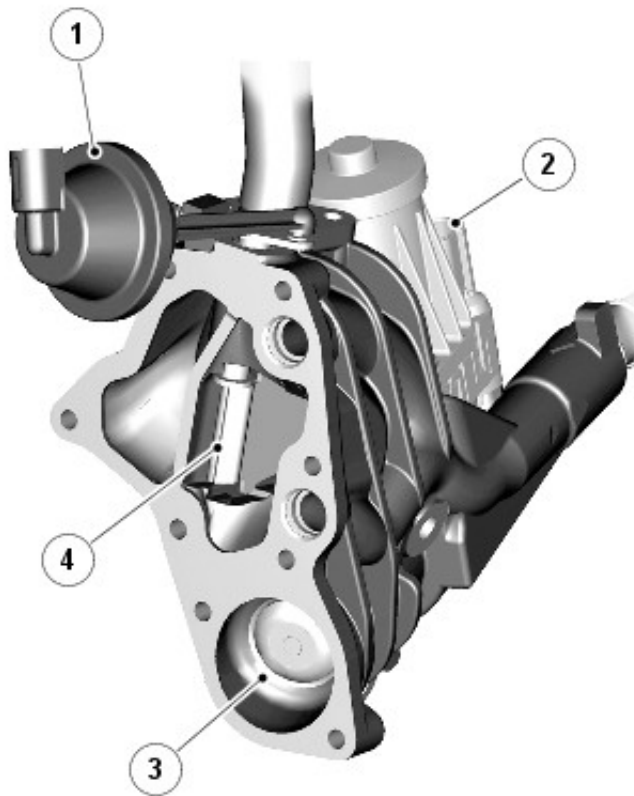
3	Vacuum connection
4	EGR intake tube
5	Cooler
6	Valve housing
7	Coolant intake connection
8	Electrical connector
9	EGR valve motor
10	EGR outlet tube connection
11	Coolant outlet connection

The EGR system comprises an EGR cooler and valve housing assembly bolted to each cylinder head, above the exhaust manifolds. Each EGR cooler and valve housing assembly incorporates an EGR valve and motor, an EGR cooler, and a cooler bypass valve and actuator.

On each valve housing, an EGR intake tube is connected to the exhaust manifold and directs exhaust gasses into the valve housing. A second pipe from the valve housing connects to the electric throttle and passes the exhaust gasses into the electric throttle to mix with the clean air entering the engine from the turbochargers.

The EGR cooler is attached to the valve housing with a gasket and 5 screws. The valve housing has an engine coolant connection which allows coolant to flow into a water jacket within the cooler, which cools the exhaust gasses. The coolant returns to the engine cooling system through an outlet pipe on the cooler.

COOLER BYPASS VALVE



E107585

ITEM	DESCRIPTION
1	Cooler bypass valve vacuum actuator
2	EGR valve motor
3	EGR valve
4	Cooler bypass valve

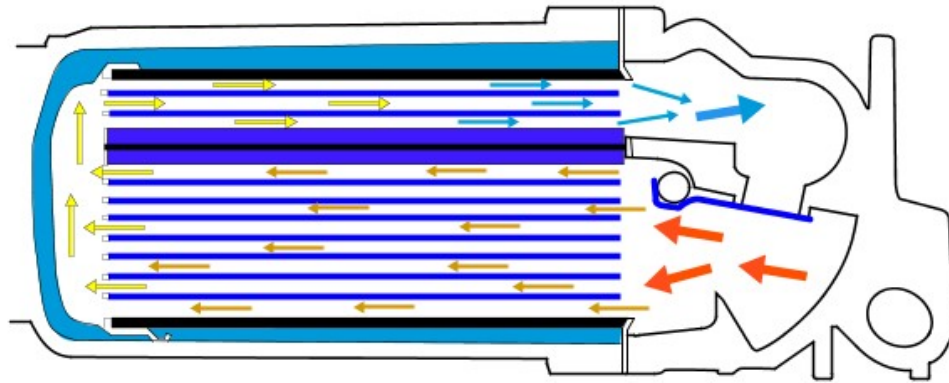
Each valve housing contains a cooler bypass valve, which either allows the exhaust gasses to flow direct to the electric throttle or diverts them through the EGR cooler.

A vacuum actuator is located on a bracket attached to each EGR cooler to operate the cooler bypass valves. The actuators receives a vacuum which is produced by the vacuum pump located at the rear of the engine. Each vacuum actuator is connected to its related bypass valve by an actuator rod.

The vacuum supply to the actuators is controlled by the ECM. When bypass

control is required, the ECM energizes a cooler bypass valve solenoid, which applies vacuum to the vacuum actuators. The vacuum causes the actuators to move the connecting rods in a linear direction. The linear movement of the rods is transferred to rotary movement of the cooler bypass valves within the valve housings.

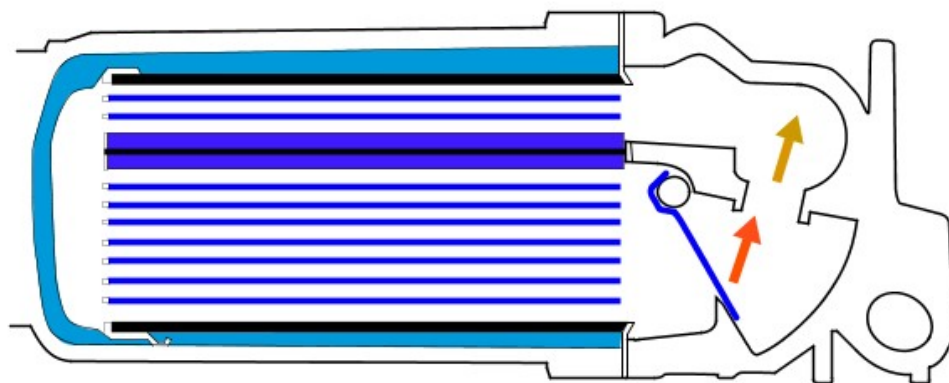
Cooler Bypass Valve Closed



E112407

When the cooler bypass valve is closed, exhaust gasses are directed through the cooler before being passed to the electric throttle.

Cooler By-Pass Valve Open



E112408

When the cooler bypass valve is open, exhaust gasses are passed directly through the valve housing into the electric throttle with no cooling applied

to the gasses.

EXHAUST GAS RECIRCULATION VALVE

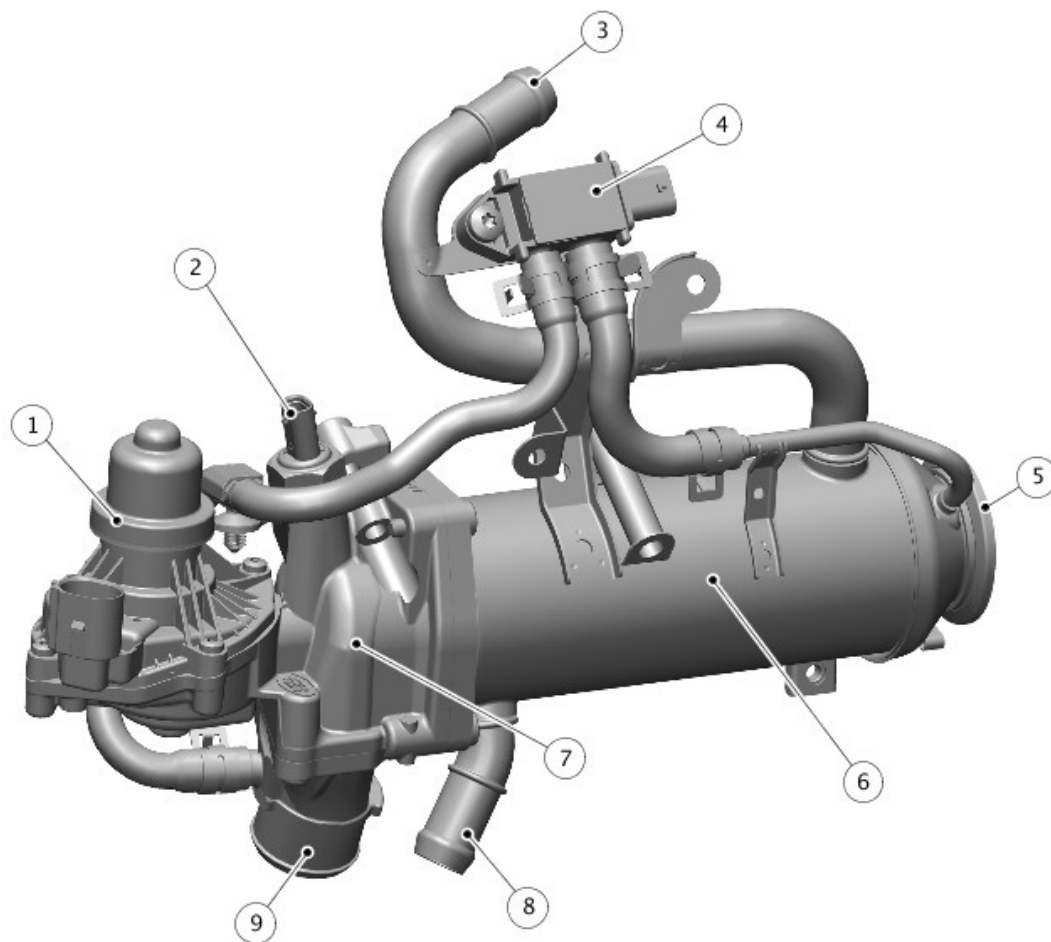
The EGR valve motor is located on the valve housing. A 5 pin connector provides the power, ground and ECM signal and feedback connections for the motor.

The motor is secured to the valve housing with 4 torx screws. A pinion gear on the motor spindle drives a geared rack which is connected to the EGR valve in the valve housing.

The motor is controlled by the ECM which provides power to operate the motor as required. A 5 Volt feedback signal is passed to the ECM which is used to establish motor position for precise control.

LOW PRESSURE (LP) EXHAUST GAS RECIRCULATION (EGR) SYSTEM

Low Pressure (LP) Exhaust Gas Recirculation (EGR)



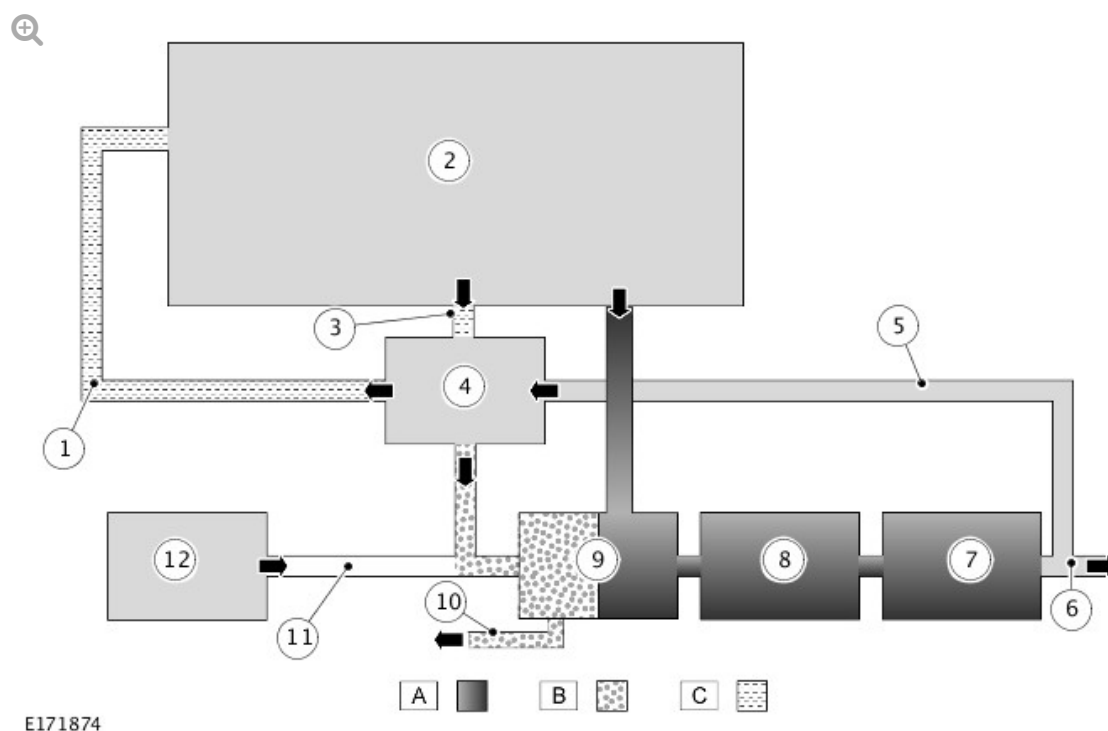
E171873

ITEM	DESCRIPTION
1	Exhaust Gas Recirculation (EGR) valve motor
2	Exhaust Gas Recirculation (EGR) outlet temperature sensor
3	Coolant inlet connection
4	Differential pressure sensor
5	Exhaust gas inlet from Post-Diesel Particulate Filter (DPF)
6	Cooler
7	Exhaust Gas Recirculation (EGR) valve housing
8	Coolant outlet connection
9	Exhaust gas outlet to clean air duct

The function of LP EGR is to re-circulate the exhaust gases after DPF (Diesel Particulate Filter) to further improve ability to meet the emission level as well as reducing the CO₂ thereby improving the fuel economy. It also improves engine transient performance due to more exhaust energy being diverted to the turbocharger(s).

As per HP EGR the function is electronically controlled and calibrated. It is governed and controlled by both hot end exhaust mounted throttle as well as unit mounted valve.

Low Pressure (LP) Exhaust Gas Recirculation (EGR) shematic diagram



A = EXHAUST GAS; B = CHARGE AIR; C = CLEAN AIR

ITEM	DESCRIPTION
1	Coolant outlet from Low Pressure (LP) Exhaust Gas Recirculation (EGR) system
2	Engine
3	Coolant inlet from engine
4	Low Pressure (LP) Exhaust Gas Recirculation (EGR) system
5	Low Pressure (LP) EGR pipe

6	Post-DPF exhaust gases
7	Diesel Particulate Filter (DPF)
8	Catalytic converter
9	Turbocharger
10	Charge air outlet
11	Clean air form Air intake system
12	Air intake system

CRANKCASE VENTILATION SYSTEM

The crankcase ventilation system comprises an oil separator and associated ventilation hoses and tubes.

The oil separator is connected to the top of the cylinder block with two seals. An oil outlet connection locates in the high pressure fuel pump mounting bracket. Two ventilation tubes connect the top of the oil separator to the camshaft covers. A ventilation tube and hose assembly connects the top of the oil separator to the intake duct of the primary turbocharger.

OPERATION

EXHAUST GAS RECIRCULATION SYSTEM

If small volumes of fuel are injected into a combustion chamber full of pure air, the effect is to create a very lean mixture. This burns at a high temperature, which in turn causes the excess oxygen in the mixture to combine with the naturally occurring nitrogen in the air to create NOX (Oxides of Nitrogen), a noxious class of pollutants associated with acid rain. This is a particular problem for diesel engines at low to medium loads. Exhaust gas is blended into the intake air charge to create the cylinder charge. As the exhaust gas effectively contains no oxygen, it prevents the formation of a very lean mixture, so lowering combustion temperatures and minimizing the formation of NOX.

At low engine speeds and loads, over 50 percent of the cylinder charge can be made up of recycled exhaust gas. This is routed from the exhaust manifolds through the EGR coolers before being supplied to the electric throttle. The volume of exhaust gas added to the intake charge is regulated by the EGR valves, actuated according to precise engine speed and load maps by the ECM.

The exhaust gases are passed through the EGR cooler before being passed through the EGR pipe to the intake manifold.

HIGH PRESSURE (HP) EXHAUST GAS RECIRCULATION SYSTEM

NOTE:

Twin HP EGR (GEN 1.5) system at 16MY is used on EU4/RoW derivatives.

The HP EGR valve is attached to the intake manifold. Exhaust gases are routed to the EGR valve from the exhaust manifold.

The EGR valve is cooled using engine coolant in the vehicle interior heating circuit; this is to protect the electrical components from over-heating.

The EGR valve controls the amount of EGR exhaust gas flow depending on the ECM map.

The exhaust gases are passed directly into the intake manifold via a simple EGR pipe on EU5 and EU6 market vehicles.

The design of the EGR pipe attachment to the inlet manifold helps mix the exhaust gases with the main fresh air from the intake air system and also insulates the plastic intake manifold from the hot EGR pipe.

On EU4 market vehicles, a separate EGR pipe attachment is used to connect between the intake manifold and the EGR cooler.

LOW PRESSURE (LP) EXHAUST GAS RECIRCULATION SYSTEM

NOTE:

HP/LP EGR (GEN 2) system is used from 15.5MY on EU5/EU6 derivatives.

The LP EGR system is used on EU5 and EU6 market vehicles only.

The LP system takes exhaust gases from the exit of the Diesel Particulate Filter (DPF) system, and mixes it with the fresh air from the intake air system into the turbocharger.

The gases are transported from the DPF outlet to the LP EGR cooler via a pipe. The LP cooler has no bypass mode, so only cools.

There is a simple mesh filter fitted before the gases reach the outlet of the cooler that prevents larger particles of soot etc. from entering the turbocharger system.

The cooled gases are then directed through the LP EGR valve.

The valve consists of a tube between the fresh air duct and the turbocharger, with the butterfly valve covering the EGR inlet, in the centre bottom of the EGR valve.

The butterfly valve opens into the fresh air stream, promoting both thorough mixing before the gas enters the turbocharger, and also provides 'suction' to drive the LP EGR gas through the system.

CRANKCASE VENTILATION SYSTEM

Clean air being drawn into the primary turbocharger when the engine is running creates a vacuum in the ventilation tube and hose assembly. This vacuum in turn creates a vacuum in the oil separator and draws gasses from the crankcase and camshaft covers into the separator. The gasses are circulated around the oil separator, allowing oil particles to be separated from the gas. The gasses are drawn into the ventilation tube and hose

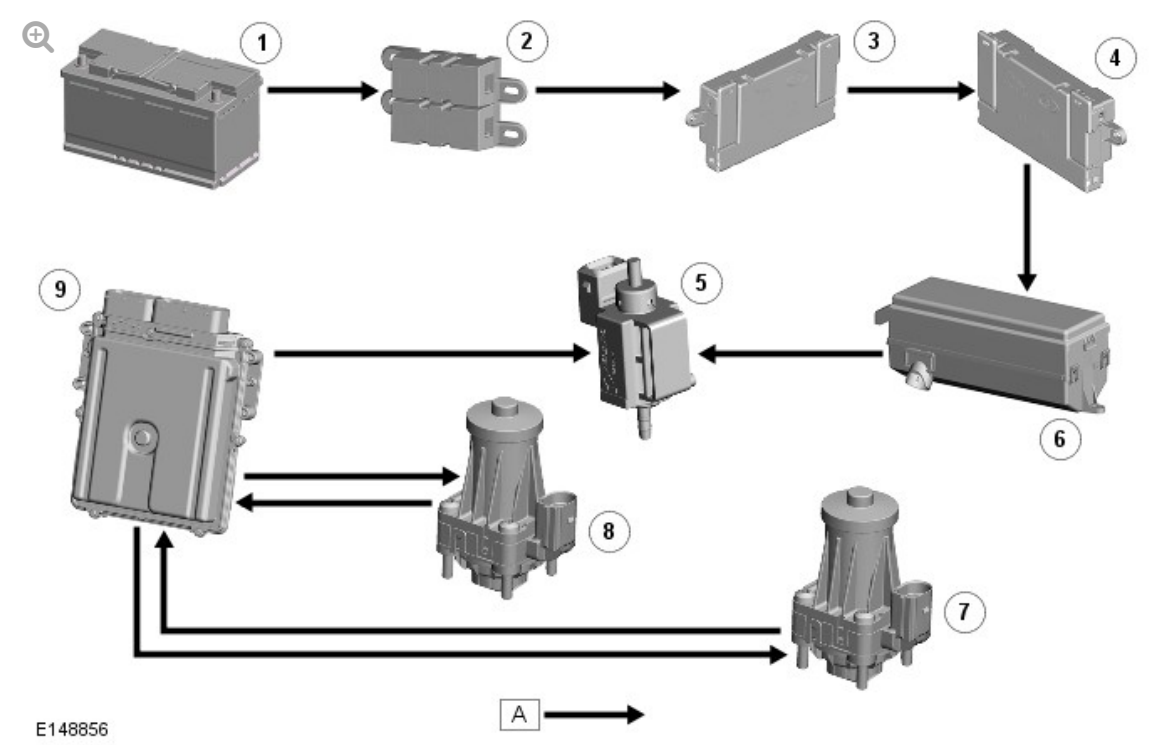
assembly and are mixed with the clean air being drawn into the turbocharger. The oil drains out of the oil separator, and down through the high pressure fuel pump mounting bracket and cylinder block into the oil pan.

DIFFERENTIAL PRESSURE SENSOR

The LP EGR differential pressure sensor monitors the pressure across the LP EGR cooler to detect issues such as cooler or gauze filter (between valve and cooler) reduced flow/blockage.

A Malfunction Indicator Lamp (MIL) will be illuminated on the Instrument Cluster (IC) if an issue is detected.

CONTROL DIAGRAM



A = HARDWIRED.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB)2
3	Battery Junction Box (BJB)

4	Auxiliary Junction Box (AJB)
5	EGR cooler bypass valve solenoid
6	Engine Junction Box (EJB)
7	Left EGR valve motor
8	Right EGR valve motor
9	Engine Control Module (ECM)